```
from google.colab import files
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Upload image
uploaded = files.upload()

# Read uploaded image
filename = list(uploaded.keys())[0]
image = cv2.imread(filename)
print("Image uploaded successfully:", filename)

# Convert image to grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Apply Sobel X to detect vertical edges
sobel_x = cv2.Sobel(
    gray, cv2.CV_64F, 1, 0, ksize=3
)

# Apply Sobel Y to detect horizontal edges
sobel_y = cv2.Sobel(
    gray, cv2.CV_64F, 0, 1, ksize=3
)

# Convert results to 8-bit images
sobel_x = cv2.convertScaleAbs(sobel_x)
sobel_y = cv2.convertScaleAbs(sobel_y)

# Combine Sobel X and Sobel Y
sobel_combined = cv2.addWeighted(
    sobel_x, 0.5,
    sobel_y, 0.5,
    0
)

# Display results
plt.figure(figsize=(12, 8))

plt.subplot(2, 2, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis("off")

plt.subplot(2, 2, 2)
plt.imshow(sobel_x, cmap="gray")
plt.title("Sobel X - Vertical Edges")
plt.axis("off")

plt.subplot(2, 2, 3)
plt.imshow(sobel_y, cmap="gray")
plt.title("Sobel Y - Horizontal Edges")
plt.axis("off")

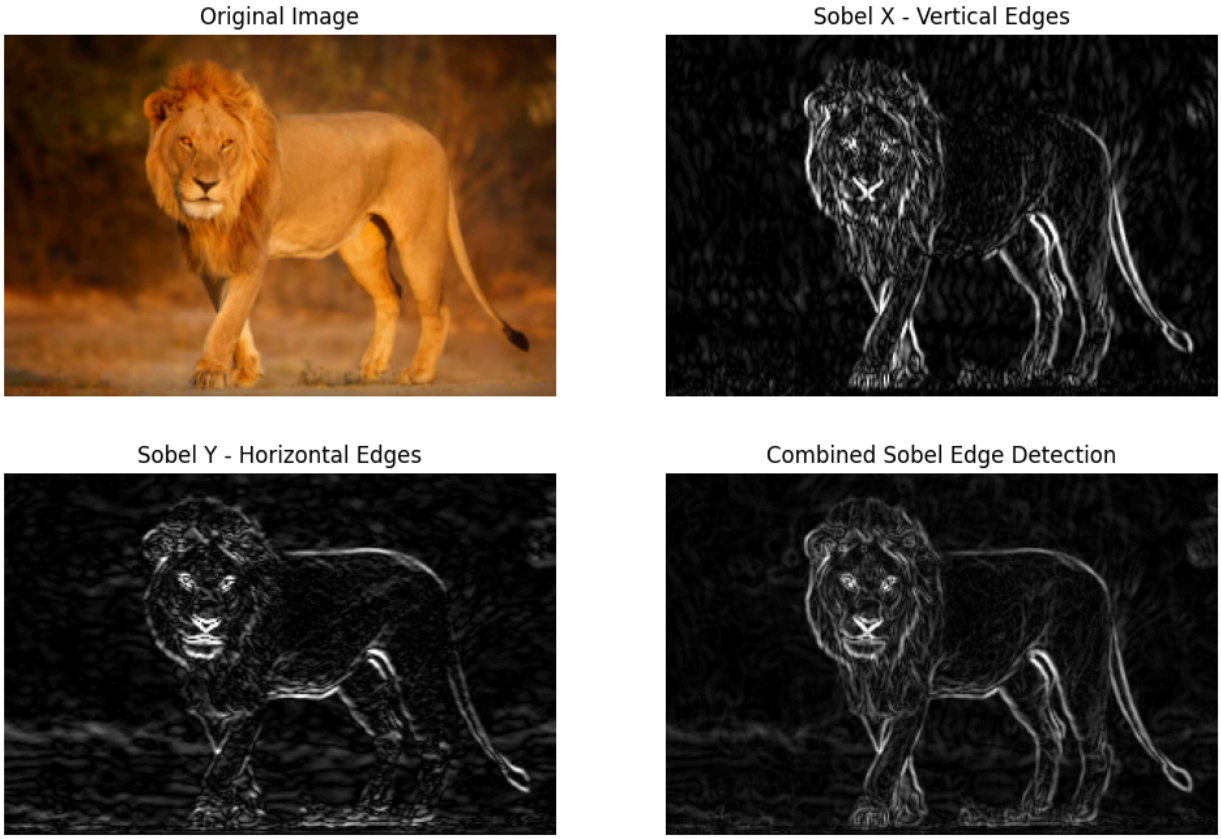
plt.subplot(2, 2, 4)
plt.imshow(sobel_combined, cmap="gray")
plt.title("Combined Sobel Edge Detection")
plt.axis("off")

plt.show()
```

Choose Files Screenshot… 075823.png

Screenshot 2026-10-10 075823.png(image/png) - 93416 bytes, last modified: 10/10/2026 - 100% done

```
Saving Screenshot 2026-10-10 075823.png to Screenshot 2026-10-10 075823.png
Image uploaded successfully: Screenshot 2026-10-10 075823.png
```



What is Colab?

Colab, or "Colaboratory", allows you to write and execute Python in your browser, with

- Zero configuration required
- Access to GPUs free of charge
- Easy sharing

Whether you're a **student**, a **data scientist** or an **AI researcher**, Colab can make your work easier. Watch Introduction to Colab or Colab Features You May Have Missed to learn more, or just get started below!

Getting started

The document you are reading is not a static web page, but an interactive environment called a **Colab notebook** that lets you write and execute code.

For example, here is a **code cell** with a short Python script that computes a value, stores it in a variable, and prints the result:

```
seconds_in_a_day = 24 * 60 * 60
seconds_in_a_day
```

```
86400
```

To execute the code in the above cell, select it with a click and then either press the play button to the left of the code, or use the keyboard shortcut "Command/Ctrl+Enter". To edit the code, just click the cell and start editing.

Variables that you define in one cell can later be used in other cells: